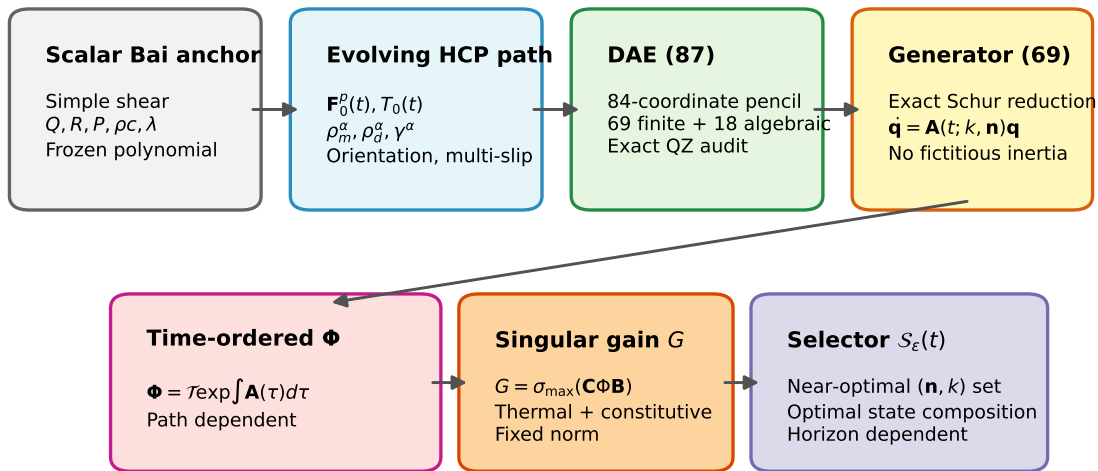




Classical theory cannot determine

which crystal perturbation accumulates the largest gain, or why direction, wavelength, and state composition change with horizon.

Finite-time crystal theory resolves

history-conditioned gain and a tested near-optimal direction--wavenumber set; state composition is conditional on norm, search box, and validation gates.